



Reconstruction of Nosocomial Infectious Disease Outbreaks Using Bayesian Inference

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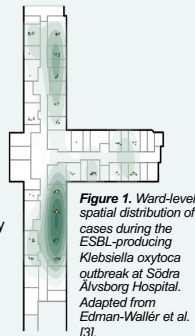
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Background

The challenge. Hospital outbreaks pose a substantial threat to patient safety and are difficult to control because transmission is only partially observed. Undetected infections and patient movement create considerable uncertainty regarding transmission pathways.

Motivation. A hospital-wide outbreak of ESBL-producing *Klebsiella oxytoca* at Södra Älvsborg Hospital [3] motivates the development of transmission-reconstruction methods. Existing outbreak-reconstruction methods, such as outbreaker2 [1], have been successfully applied to reconstruct transmission chains in partially observed outbreaks [2], but do not explicitly account for hierarchical hospital structure.

Goal. Develop a Bayesian framework for reconstructing transmission chains in structured hospital environments



Epidemic Properties

SIS model. Patient turnover generates SIS-like dynamics.



Basic reproduction number:

$$R_0 = \frac{\beta_R n_R}{p_d} + \frac{\beta_W n_W}{p_d} + \frac{\beta_H n_H}{p_d}$$

$$R_0 = R_R + R_W + R_H$$

Endemicity: $R_0 \leq 1 \Rightarrow$ extinction

$$R_0 > 1 \Rightarrow$$
 persistence

Endemic prevalence: $I^* \approx 1 - \frac{1}{R_0}$

Threshold and prevalence follow standard epidemic theory [4].

Model

Discrete-time stochastic transmission model

Transmission rates:

$$\beta_H < \beta_W < \beta_R$$

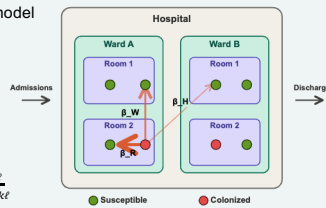
Daily probability of infection:

$$P_t(k) = 1 - \prod_{\ell \in I(t)} (1 - \beta_{S_\ell}(k, \ell))$$

where $S_\ell(k, \ell) \in \{H, W, R\}$

Ancestor attribution: $P(\alpha_k = \ell) \propto \frac{\beta_{k\ell}}{1 - \beta_{k\ell}}$

Discharge, testing, mutation rates: p_d, p_t, μ



Example reconstruction (synthetic outbreak)

Synthetic outbreak (2 wards, 6 rooms, 18 beds, 45 days) reconstructed from spatial, temporal, and genetic data.

Legend:

- Node = observed case
- Bar = hospital stay
- Edge = transmission
- Tick = test date
- X-axis = days
- Y-axis = beds

Performance:

- MAP accuracy: 61%
- True-ancestor posterior probability: 58%

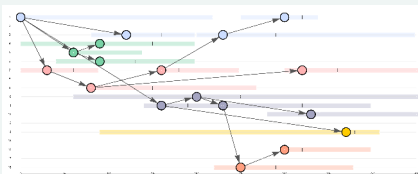


Figure 2. True transmission tree.



Figure 3. MAP reconstruction. Green edges are correct; red edges are incorrect.

Discussion & Conclusion

Conclusion. Validation on synthetic hospital outbreaks demonstrates that accurate reconstruction of nosocomial transmission pathways is possible despite incomplete surveillance, with MAP ancestor-attribution accuracies approaching 75–80%. Because the required data are routinely collected in hospitals, Bayesian outbreak reconstruction could become a valuable tool for infection prevention and control.

Future work:

- Application to the motivating SÄS outbreak dataset.
- Integration of environmental reservoirs into both simulation and inference.
- Further methodological development of transmission-tree likelihoods and MCMC sampling schemes.

Outbreak Reconstruction

Infer who infected whom from partial observations.

Observed outbreaks are incomplete realizations of a latent transmission network.

Observed data: $D = (T_a, T_d, T_t, S, C)$

- T_a : Admission date
- T_d : Discharge date
- T_t : Test date
- s : Pathogen genome
- C : Patient location

Latent variables: $A = (\alpha, \kappa, T_i)$

- α : Ancestor
- κ : Number of hidden intermediate cases
- T_i : Infection time

Posterior distribution: $P(A, \theta | D) \propto P(A, D | \theta) P(\theta)$

• Joint inference of transmission history and model parameters

Likelihood: Temporal • Spatial • Testing • Genetic

Inference: Sample from posterior using Metropolis-within-Gibbs MCMC.

- Update infection times T_i
- Reassign ancestors α among eligible cases
- Update number of hidden intermediate cases κ
- Occasional joint (α, T_i) move for improved mixing

Validation

Setup:

- 35 simulated outbreaks per testing rate
- 2000 MCMC iterations per outbreak
- MAP transmission tree used for evaluation

Key findings:

- Accuracy approaches 75–80%
- Limited sensitivity to R_0

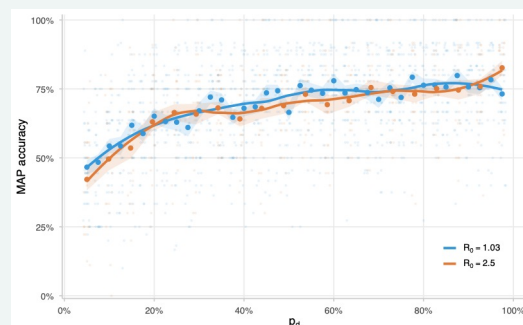


Figure 4. MAP ancestor-attribution accuracy across synthetic outbreaks as a function of testing rate. Colors indicate outbreaks with $R_0 = 1.03$ and $R_0 = 2.5$; points show individual replicates and lines show smoothed trends.

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- [2] Duault H, Durand B, Canini L (2024). Outbreak reconstruction with a slowly evolving multi-host pathogen: a comparative study of three existing methods on *Mycobacterium bovis* outbreaks. *Epidemics* 49:100794.
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